

Beyond Constant Drift: Comparison-Function Exit Certificates

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September 23, 2026

Abstract

The timing certificates of the obstruction calculus assume constant adverse drift; resource systems rarely decline linearly. This paper, the D9 item of the generalization catalogue, extends the exit certificate along the catalogue's three axes and verifies each as an exact identity on the programme's grids. State-dependent drift: with per-step comparison function α , the last safe step is governed by the comparison integral, and for $\alpha(q) = q/10$ the exit value recovers the successor's $\sigma^*(q_0) = \max\{k : q_0 \geq (\frac{10}{9})^k\}$ exactly at the powers — the constant-drift theory embeds into the comparison-function theory without remainder on the contraction instance. Nonlinear comparison: for $\alpha(q) = q^2/10$ the integral bound $10(1 - 1/q_0)$ dominates the exact iteration exit time on the whole grid, strictly off the boundary. Multi-constraint exit: with $q_{\min} = \min_j q_j$ the last safe step of the jointly declining two-floor system equals $\lfloor 10 q_{\min}(0) \rfloor$ on the 240-point grid — “at least one floor breached” certified by the minimum-slack drift. A scope instance closes the paper honestly: when α vanishes at the boundary ($\alpha(q) = q/10 - \frac{1}{20}$ at $q = \frac{1}{2}$) the iteration converges to the equilibrium and never exits — no finite bound exists and the comparison integral diverges; positivity of the comparison function at the boundary is load-bearing (6/6 check families).

Introduction and relation to the programme

Paper 2's finite-time exit certificate supposes $\sup_u \inf_d D^+q \leq -\varepsilon$; the catalogue proposes the three generalizations adopted here — state-dependent drift, comparison functions, and multi-constraint exit through the minimum slack — noting that they make the timing layer applicable where decline accelerates near thresholds. The successor's exit-time values supply the object whose level sets these bounds reproduce; nothing in the completed items is altered.

Comparison-function exit bounds

Let the floor margin $q \in (0, a]$ follow the adversarial one-step law $q_{t+1} \leq q_t - \alpha(q_t)$ with α positive on $(0, a]$. Then

the last safe step is bounded by the comparison integral:

$$t_{\text{exit}} \leq \int_{q_{\text{thr}}}^{q_0} \frac{dq}{\alpha(q)}.$$

Proposition 1 (Contraction embedding). For $\alpha(q) = q/10$ (per-step multiplicative decline $q_{t+1} = (\frac{9}{10})q_t$), the last safe step below 1 equals $\max\{k : q_0 \geq (\frac{10}{9})^k\}$ — the successor's exit-time value — with equality at the exact powers $(\frac{10}{9})^K$ and the recorded neighbors. The constant-drift theory is the $\alpha(q) = \varepsilon$ case.

Proof. The iterate $q_k = q_0(\frac{9}{10})^k$ satisfies $q_k \geq 1$ iff $q_0 \geq (\frac{10}{9})^k$; taking the maximum over k is the successor's value. The constant case is the arithmetic-progression analogue with the floor identity of the verified record. $\square$

Proposition 2 (Dominance and monotonicity). For $\alpha(q) = q^2/10$ the bound $10(1 - 1/q_0)$ dominates the exact iteration exit time on the grid (strictly off the boundary), and larger comparison functions give tighter bounds: the bound under $\alpha_2 = q/10$ is at most (strictly, off the boundary) the bound under $\alpha_1 = q/20$.

Proof. The integral $\int_1^{q_0} 10 q^{-2} dq = 10(1 - 1/q_0)$ upper-bounds the steps of any path obeying the per-step decline; pointwise dominance of $1/\alpha$ gives the monotonicity. $\square$

Proposition 3 (Multi-constraint exit). With $q_{\min} = \min_j q_j$ and joint decline $z'_j = z_j - \varepsilon$, the system exits $\mathcal{V}$ exactly when $q_{\min} < 0$, and the last safe step equals $\lfloor q_{\min}(0)/\varepsilon \rfloor$: a drift condition on the minimum slack certifies that at least one floor is breached, within a computed budget.

Proof. $\min_j$ of equal declines declines equally; the exit event of the vector system is the negativity of the minimum slack; the identity is the constant-drift arithmetic in $q_{\min}$. $\square$

Verified record (6/6 exact check families)

E1 the constant-drift identity (last safe step = $\lfloor 10(z_0 - 1) \rfloor$, 48 cells); E2 the contraction embedding at exact powers and $\frac{1}{100}$ -neighbors ($K \leq 3$); E3 the quadratic-comparison dominance on the grid with strict domination recorded; E4 the $q_{\min}$ identity on the 240-point grid; E5 comparison monotonicity (α_2 vs α_1); E6 the scope instance: $\alpha(q) = q/10 - \frac{1}{20}$ vanishes at $q = \frac{1}{2}$, the iteration converges there from above and never exits, the integral diverges — no finite bound exists, and the naive application of the bound is not merely loose but undefined.

What is not claimed

The comparison-function layer is one-dimensional in the floor margin: no vector-valued comparison functions, no stochastic comparison theorems (the stochastic selector’s setting), and no converse — a finite integral bound is sufficient for exit, and nothing is claimed about necessity. The audited instances are discrete one-step iterations of the programme’s working systems; continuous-time flows are cited through the successor’s HJB/Isaacs layer, not developed. The scope instance is a property of the affine example, not a classification of all boundary-vanishing comparison functions.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `comparison_drift_exit_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.